

Sacha Morin

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Education

- 2022–present **PhD, Computer Science**, Université de Montréal - Mila
Machine Learning and Robotics
Expected Graduation Date: 08/2027
Advisors: Guy Wolf and Liam Paull
Labs:
○ Mila - Quebec Artificial Intelligence Institute
○ RAFALES
○ Robotics and Embodied AI Lab (REAL)
- 2021–2022 : **Research MSc, Computer Science**, Université de Montréal - Mila
Machine Learning. Not completed. Fast tracked to PhD.
Advisor: Guy Wolf
GPA: 4.30/4.30
- 2017–2021 : **Bachelor of Mathematics and Computer Science**, Université de Montréal
GPA: 4.15/4.30
- 2014–2017 : **Bachelor of Law (LLB)**, Université de Sherbrooke
GPA: 3.68/4.30

Research Experience

Mila - Quebec AI Institute

- 2022–present **Phd Student**, *Montreal, QC, Canada*
○ 3D multimodal scene representations for robotics and embodied AI [C1], [B1], [P1], [C2].
○ Representation learning for robotics [C3], [C4].
○ Robot learning.
Advisors: Guy Wolf and Liam Paull
- 2020–2022 **Research Intern/MSc Student**, *Montreal, QC, Canada*
○ Learning structured and interpretable representations by combining autoencoders and manifold learning [J3], [C5].
○ Apply clustering and data visualization tools for unsupervised exploration of biological datasets [J2], [P3], [J4].
Advisor: Guy Wolf

Samsung's SAIT AI Lab (SAIL)

March 2025 - **Research Intern**, *Montreal, QC, Canada*

- Sept 2025 ○ Joint video and policy learning.
- Latent action modeling.

Advisors: Alexia Jolicoeur-Martineau & Sébastien Lachapelle

Nokia Bell Labs

Summer 2024 **Research Intern**, *Murray Hill, NJ, United States*

- 3D perception with mobile robots.
- LLM integration with multimodal 3D representations.
- Topological language mapping.

Advisor: Dan Kushnir

Université de Montréal

2018-2019 **Research Intern**, *Montreal, QC, Canada*

- Develop Gambit Forensics, an analytics tool to benchmark various compilers of the Scheme programming language.

Advisor: Marc Feeley

Work Experience

LJT Laywers LLP

Jan 2020- **Lawyer (Part Time)**, *Montreal, QC, Canada*

April 2020 Mergers and acquisitions of software companies. Contract Law.

June 2019- **Articling Student**, *Montreal, QC, Canada*

Jan 2020 Mergers and acquisitions of software companies. Contract Law. Litigation.

Supervisor: Me Nicolas Lassonde

May 2018- **Law Student**, *Montreal, QC, Canada*

Aug 2018 Contract Law. Litigation.

Training

2023 **ETH Zurich Robotics Summer School**, *Avully, Switzerland*

- Search and rescue with mobile robots. Our team won the competition!

2022 **Mila Robotics Summer School**, *Montreal, QC, Canada*

- Introduction to quadruped robots and RL.

Awards

2023 **NSERC PGS D Scholarship**, Natural Sciences and Engineering Council of Canada

2023 **FRQNT Doctoral Scholarship**, Fonds de recherche du Québec - Nature et technologies

2021 **IVADO M.Sc. Scholarship**, Institut de valorisation des données

2021 **FRQNT B1X Scholarship**, Fonds de recherche du Québec - Nature et technologies

2021 **NSERC M.Sc. Scholarship (Declined)**, Natural Sciences and Engineering Council of Canada

- 2021 **ISM Undergraduate Research Scholarship**, Institut des sciences mathématiques
- 2020 **IVADO Undergraduate Research Scholarship**, Institut de valorisation des données
- 2019 **NSERC Undergraduate Award**, Natural Sciences and Engineering Research Council of Canada
- 2019 **Scholarship for Academic Excellence**, Bourse d'excellence des diplômés et des professeurs
- 2016 **Winner of the Matthieu-Bernard Competition**, Société québécoise de droit international
- 2015-2017 **Dean's List**, Faculty of Law, Université de Sherbrooke

Teaching and Academic Involvement

- 2023-2025 **Teaching Assistant**, STT 3795: Theoretical Foundations of Data Science, Université de Montréal
 - Undergraduate class taught by Prof. Guy Wolf. Gave some lectures.
- 2023-2024 **Organizer**, Robot Learning Seminar, Mila - Quebec AI institute
 - [YouTube Playlist](#)
- 2023 **Co-organizer**, Mila Robotics Summer School, Mila - Quebec AI Institute
 - Prepare workshop and challenge using the Unitree Go1 robot and TagSLAM.
- 2023 **Volunteer**, Conference on Robots and Vision (CRV), Montreal, Quebec, Canada
- 2022-2023 **Member**, IVADO Student Intersectoral Committee
 - Support major IVADO events, such as job fairs and *Digital October*.
- 2022 **Invited Talk on AI, Data & Algorithms**, Prof. Sylvano Santini's SEM9500 Seminar, Université du Québec à Montréal
- 2014-2015 **Pro Bono Canada**, Université de Sherbrooke
 - Draft training material for directors of non-profits.

Reviewer

CoRL 2025, ICCV 2025, IROS 2025, RA-L 2024, RSS 2024, T-RO 2023, IROS 2023, ICRA 2023

Software

- 2023 **StepMix**, A Python package following the scikit-learn API for model-based clustering and generalized mixture modeling of continuous and categorical data [[J1](#)].

Skills

- Programming Languages Python. C++. Some knowledge of C, R, JAVA and Javascript.
- Libraries PyTorch, ROS, Scikit-Learn, Pandas, NumPy.

Publications & Preprints

* indicates joint authorship.

Book Chapters

- [B1] L. Paull, **S. Morin**, D. Maggio, M. Büchner, C. Cadena, A. Valada, and L. Carlone, “Towards open-world spatial ai”, in *SLAM Handbook*, To appear, 2025.

Journal Publications

- [J1] **S. Morin***, R. Legault*, F. Laliberté, Z. Bakk, C.-É. Giguère, R. de la Sablonnière, and É. Lacourse, “Stepmix: A python package for pseudo-likelihood estimation of generalized mixture models with external variables”, *Journal of Statistical Software*, vol. 113, pp. 1–39, 2025.
- [J2] E. Brunet-Ratnasingham*, **S. Morin***, H. E. Randolph*, M. Labrecque, J. Bélair, R. Lima-Barbosa, A. Pagliuzza, L. Marchitto, M. Hultström, J. Niessl, *et al.*, “Sustained ifn signaling is associated with delayed development of sars-cov-2-specific immunity”, *Nature Communications*, vol. 15, no. 1, p. 4177, 2024.
- [J3] A. F. Duque*, **S. Morin***, G. Wolf, and K. R. Moon, “Geometry regularized autoencoders”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.
- [J4] B. Paré, M. Rozendaal, **S. Morin**, L. Kaufmann, S. M. Simpson, R. Poujol, F. Mostefai, J.-C. Grenier, H. Xing, M. Sanchez, *et al.*, “Patient health records and whole viral genomes from an early SARS-CoV-2 outbreak in a Quebec hospital reveal features associated with favorable outcomes”, *Plos one*, vol. 16, no. 12, e0260714, 2021.

Conference Proceedings

- [C1] C. Kassab*, **S. Morin***, M. Büchner*, M. Mattamala, K. Gupta, A. Valada, L. Paull, and M. Fallon, “Openlex3d: A new evaluation benchmark for open-vocabulary 3d scene representations”, in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, To appear, 2025.
- [C2] Q. Gu*, A. Kuwajerwala*, **S. Morin***, K. Jatavallabhula*, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. de Melo, J. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, “Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning”, *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [C3] **S. Morin***, M. Saavedra-Ruiz*, and L. Paull, “One-4-All: Neural potential fields for embodied navigation”, in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023.
- [C4] M. Saavedra-Ruiz*, **S. Morin***, and L. Paull, “Monocular robot navigation with self-supervised pretrained vision transformers”, in *2022 19th Conference on Robots and Vision (CRV)*, IEEE, 2022, pp. 197–204.
- [C5] A. F. Duque*, **S. Morin***, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *2020 IEEE International Conference on Big Data (Big Data)*, IEEE, 2020, pp. 5027–5036.

Preprints

- [P1] **S. Morin**, K. Gupta, M. Sandhu, C. Gauthier, F. Argenziano, K. Ellis, and L. Paull, “Agentic scene policies: Unifying space, semantics, and affordances for robot action”, *arXiv preprint arXiv:2509.19571*, 2025.
- [P2] C. Kassab, M. Mattamala, **S. Morin**, M. Büchner, A. Valada, L. Paull, and M. Fallon, “The bare necessities: Designing simple, effective open-vocabulary scene graphs”, *arXiv preprint arXiv:2412.01539*, 2024.
- [P3] J. S. Rhodes, A. Aumon, **S. Morin**, M. Girard, C. Larochelle, E. Brunet-Ratnasingham, A. Pagliuzza, L. Marchitto, W. Zhang, A. Cutler, *et al.*, “Gaining biological insights through supervised data visualization”, *bioRxiv*, pp. 2023–11, 2023.

Workshops

- [W1] F. Argenziano, M. Saavedra-Ruiz, **S. Morin**, D. Nardi, and L. Paull, “Dynamic objects relocalization in changing environments with flow matching”, in *Perception and Planning for Mobile Manipulation in Changing Environments @ IROS*, 2025.
- [W2] **S. Morin**, K. Gupta, M. Sandhu, C. Gauthier, F. Argenziano, K. Ellis, and L. Paull, “Agentic scene policies: Unifying space, semantics, and affordances for robot action”, in *NeurIPS 2025 Workshop on Space in Vision, Language, and Embodied AI (SpaVLE)*, 2025.
- [W3] **S. Morin***, S. Naht*, S. Ebrahimi Kahou, and G. Wolf, “Spectral temporal contrastive learning”, in *NeurIPS 2023 Workshop: Self-Supervised Learning - Theory and Practice*, 2023.
- [W4] A. F. Duque*, **S. Morin***, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *NeurIPS 2020 Workshop on Differential Geometry Meets Deep Learning (DiffGeo4DL)*, 2020.
- [W5] A. F. Duque*, **S. Morin***, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *DeepMath 2020 Conference on the Mathematical Theory of Deep Neural Networks*, 2020.
- [W6] **S. Morin***, A. F. Duque*, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *Montreal AI Symposium (MAIS)*, 2020.

Updated September 2025.